

PSYCH 391MB: Seminar - Brains, Minds, and Machines

TuTh 10:10AM - 11:15AM

Tobin Hall 520

Instructor: Dr. Lauren Aulet

Office Hours: Wednesdays at 3-4pm, or by appointment (Tobin 425)

Summary: Brains, Minds, and Machines explores how biological and artificial systems represent information, learn from experience, and develop specialized abilities. Across the semester, we will ask what counts as a representation, why brains and neural networks become specialized, what must be built into a learner versus learned from experience, when a neural network can meaningfully serve as a model of cognition, and ultimately what we mean by “intelligence.” We will approach these questions by comparing brains, developing minds, and artificial neural networks, with particular emphasis on visual cognition, learning, and intelligence.

Readings: All readings will be supplied on Canvas.

Note: No class 10/13 (Indigenous Peoples’ Day, Mon schedule followed on Tue), 11/3 (Election Day), or 11/26 (Thanksgiving recess).

Date	Topic	Reading / Material
9/8	Syllabus and Background	—
9/10	Neuroanatomy crash course	—
9/15	Neural methods crash course	—
9/17	Neural Networks crash course	3Blue1Brown Neural Networks series: https://www.youtube.com/watch?v=aircAruvnKk
9/22	NNs as models of the brain	Lindsay 2020
9/24	Transformers and LLMs crash course	3Blue1Brown LLM video: https://youtu.be/wjZofJX0v4M?si=mhKOHAv6gFzl5lvM
9/29	NNs as models of development	Smith & Slone 2017

10/1	NNs as models of development	Ayzenberg et al. 2026
10/6	Object recognition	Ayzenberg 2025
10/8	Localization/modularity in the brain	Pitcher 2009; Reflection paper #1 due
10/15	Localization/modularity in DNNs	Dobs 2022
10/20	Face recognition	Nancy Kanwisher MIT talk https://nancysbraintalks.mit.edu/video/what-you-can-learn-studying-behavior
10/22	Word/Letter & Number recognition	Dehaene 2007; Hannagan 2015
10/27	Extreme specialization / localist representations	Connor 2005; https://www.anthropic.com/research/mapping-mind-language-model
10/29	Against specialization: distributed and mixed representations	Lashley 1950; https://www.nytimes.com/2022/09/04/science/brain-language-research.html
11/5	Open discussion / catch up	Reflection paper #2 due
11/10	History of cognitive science	Tolman 1948
11/12	Nativism and inductive biases in humans and machines	Spelke & Kinzler 2007
11/17	Statistical learning: learning structure from experience	Saffran et al. 1996
11/19	Learning mechanisms: development,	Smith 2018 TiCS

	representations, and experience	
11/24	Active learning, exploration, and learning to learn	Gopnik 2012; Xu & Kushnir 2013; Research proposal outline due
12/1	Intelligence: history, eugenics, and warnings	https://www.youtube.com/watch?v=JeCKftkNKJ0 Duyme et al., 1999
12/3	Analogical reasoning and fluid intelligence: how do we measure intelligence?	Yiu et al 2025
12/8	Machine Intelligence: ARC, then and now	https://www.nytimes.com/interactive/2025/03/26/business/ai-smarter-human-intelligence-puzzle.html
12/10	Creativity in humans and machines	Wang et al. 2026
12/15	Curiosity	Kidd & Hayden 2015; Research proposal due

Grading / Assignments:

Reflection Paper #1: 50 points

Reflection Paper #2: 50 points

Research Proposal Outline: 50 points

Research Proposal: 100 points

Final grade out of 250 points (*No extra credit via SONA systems is offered for this course.*)

Reflection paper expectations: You will complete two short reflection papers over the course of the semester (approximately 750–1,000 words each). The goal of these papers is not to summarize the readings, but to develop a substantive reaction to something we have discussed in class. Identify an idea, argument, result, or tension that you found interesting and think through it further. You might challenge a claim or interpretation, connect ideas across readings, propose an alternative explanation, identify an unresolved question, or consider a broader implication of the work (or something else entirely, I'm open to creativity with these!). I am primarily looking for evidence of your own thinking rather than comprehensive coverage of the material. I'd rather read 750 words containing one interesting idea than

1,500 words summarizing six papers. You are welcome to use AI tools in preparing these assignments, including for brainstorming, organization, or writing assistance, but the central insight, argument, or perspective should be genuinely yours, and you are responsible for the accuracy and content of everything you submit.

Research proposal expectations: Your final assignment will be a research proposal in which you develop an original study related to one or more of the questions we have considered in this course. Proposals should be approximately 6–10 single-spaced pages, plus references, although I care much more about whether the proposal is complete and well reasoned than about the exact page count. Your proposal should include: (1) an introduction that reviews the relevant theoretical and empirical background and motivates the question; (2) a clear gap, unresolved question, or tension in the existing literature; (3) specific research question(s) and hypotheses; (4) a methods section describing the participants or data, stimuli/materials, study design, measures, and procedure in enough detail that the reader can understand how the study would work; and (5) a proposed analysis section explaining how the resulting data would be used to answer your research question(s). Figures or diagrams are welcome when they help communicate the design or predicted results. The goal is to demonstrate that you can take an interesting theoretical idea and turn it into a clear, testable research question and a study capable of answering it. As with the reflection papers, you are welcome to use AI tools for brainstorming, literature organization, coding/statistical questions, or writing assistance, but the central research idea and intellectual rationale should be genuinely yours, and you are responsible for understanding and defending every aspect of the proposal. I would much rather read a simple, well-motivated study that cleanly answers an interesting question than an elaborate study containing every method you have learned this semester.

Grade Scale:

A	93-100	D+	67-69
A-	90-92	D	63-66
B+	87-89	F	0-62
B	83-86		
B-	80-82		
C+	77-79		
C	73-76		
C-	70-72		

Academic Honesty:

Since UMass Amherst is strongly committed to academic integrity, which is defined as completing all academic work without cheating, lying, stealing, or receiving unauthorized assistance from any other person, or using any source of information not appropriately authorized or attributed. As a community, we hold each other accountable and support each other's knowledge and understanding of academic integrity. Academic dishonesty is prohibited in all programs of the University and includes, but is not limited to: Cheating, fabrication, plagiarism, lying, facilitating dishonesty, via analogue and digital means. Sanctions may be imposed on any student who committed or participated in an academic integrity infraction. Any person who has reason to believe that a student has committed an academic integrity infraction should bring such information to the attention of the appropriate course instructor as soon as possible. All students at the University of Massachusetts Amherst have read and acknowledged the Commitment to Academic Integrity and are knowingly responsible for completing all work with integrity and in accordance with the policy: <https://www.umass.edu/senate/book/academic-regulations-academic-integrity-policy>

Accommodation:

The University of Massachusetts Amherst is committed to providing an equal educational opportunity for all students. If you have a documented physical, psychological, or learning disability on file with Disability Services (DS), you may be eligible for reasonable academic accommodations to help you succeed in this course. If you have a documented disability that requires an accommodation, please notify me within the first two weeks of the semester so that we may make appropriate arrangements. For further information, please visit Disability Services (<https://www.umass.edu/disability/>)

Diversity:

In recognition and affirmation of the worth and dignity of all persons, the instructor of this class is dedicated to upholding the values of diversity and inclusion and to minimizing disparities in access to learning. All students who are willing to learn are equally welcome without regard to age, race, ethnicity, birthplace, sex, gender identity, disability, sexual orientation or identity, socioeconomic status, political party or beliefs, or other group affiliation or personal characteristics. All students are likewise expected to accord each other the same level of respect, bearing in mind that becoming more respectful and inclusive is a journey rather than a destination. We will all have stumbling points along the way, including the instructor, and these are often valuable learning opportunities. If I ever say or do anything insensitive in this class, I sincerely welcome students to call me out or communicate their discomfort to me in person after class, during office hours, by email, or anonymously.

Title IX:

In accordance with Title IX of the Education Amendments of 1972 that prohibits gender-based discrimination in educational settings that receive federal funds, the University of Massachusetts Amherst is committed to providing a safe learning environment for all students, free from all forms of discrimination, including sexual assault, sexual harassment, domestic violence, dating violence, stalking, and retaliation. This includes interactions in person or online through digital platforms and social media. Title IX also protects against discrimination on the basis of pregnancy, childbirth, false pregnancy, miscarriage, abortion, or related conditions, including recovery. There are resources here on campus to support you. A summary of the available Title IX resources (confidential and non-confidential) can be found at the following link: <https://www.umass.edu/titleix/resources>. You do not need to make a formal report to access them. If you need immediate support, you are not alone. Free and confidential support is available 24 hours a day / 7 days a week / 365 days a year at the SASA Hotline 413-545-0800.